CIA: model notes

Based on `CIA.mod`; Walsh (2010), Appendix 3.6, pp. 128–130.

Underlying preferences

$$U(c, 1-n) = \frac{c^{1-\eta_c}}{1-\eta_c} + \Psi \frac{(1-n)^{1-\eta_n}}{1-\eta_n}.$$

At $\eta_n = 1$, the leisure term is $\Psi \log(1-n)$. The binding cash-in-advance constraint implies $c_t = m_t$ in equilibrium. The variable λ (`lambdda`) is the marginal value of wealth, not marginal utility of consumption: the cash constraint creates a wedge.

Scope and calibration

These are stationary, log-linear business-cycle equations. Quantities are log deviations; r, i and π are level deviations of net rates. The system uses small-rate approximations, not exact gross-rate scaling. The leisure curvature implies a Frisch elasticity $(1-n^{ss})/(\eta_n n^{ss})$.

The code fixes `nss`; Ψ is implicit. Changing trend inflation while retaining `nss` changes the implied leisure weight. It does not trace steady-state hours with preferences fixed. Additive leisure preferences also generally do not support constant hours along balanced growth unless consumption-composite curvature is one; these exercises abstract from growth.

Linear equations implemented in the code

All equations are conditional on information at t; future variables in Dynare imply conditional expectations. Hats are suppressed, as in the code. Write $YK = y^{ss}/k^{ss}$, $CK = YK - \delta$ and $NL = n^{ss}/(1 - n^{ss})$.

$$
\begin{aligned}
y_t &= z_t + \alpha k_{t-1} + (1-\alpha) n_t, \\
YK\, y_t &= CK\, c_t + \delta x_t, \\
k_t &= (1-\delta) k_{t-1} + \delta x_t, \\
r_t &= \alpha YK (E_t y_{t+1} - k_t), \\
c_t &= m_t, \\
\lambda_t &= E_t \lambda_{t+1} + r_t, \\
y_t + \lambda_t &= (1 + \eta_n NL) n_t, \\
i_t &= r_t + E_t \pi_{t+1}, \\
\lambda_t &= -E_t(\eta_c m_{t+1} + \pi_{t+1}), \\
m_t &= m_{t-1} - \pi_t + \mu_t, \\
z_t &= \rho_z z_{t-1} + \epsilon_{z,t}, \\
\mu_t &= \rho_\mu \mu_{t-1} + \phi_z z_{t-1} + \epsilon_{\mu,t}.
\end{aligned}
$$

In code: $\alpha =$ `alppha`, $\beta =$ `betta`, $i =$ `int`, $\pi =$ `pii`, investment is `x`. The current calibration sets $\phi_z = 0$.